

TO VU PHUC DANG

Gameplay AI & Core Systems Engineer — Computer Engineering Student

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PROFESSIONAL SUMMARY

Computer Engineering student at VNU-UET who builds the layer where a language model's output turns into a frame-accurate gameplay decision. Comfortable on both sides of that boundary: writing the deterministic game loop, AABB collision resolution, and FSM-driven boss AI on one end, and the raw HTTP orchestration layer feeding it LLM output on the other. Entering **Prompt to Play 2026** to build a boss encounter where the AI system is load-bearing, not decorative.

TECHNICAL & GAMEPLAY AI EXPERIENCE

SOICT Innovation Club (SINNO) | *Vice Head of Incubation Department* 10/2025 – Present

- Ran the incubation pipeline for student MVP teams – pushed codebase architecture reviews and API-integration decisions instead of feature counts, cutting rework in later sprints.
- Organized and ran technical code-jams (e.g. Codegame) under hard time budgets; owned the logistics stack end-to-end, from problem-set design to on-site infra stability.

Weamis Tech Solution | *Software Developer Intern* 04/2026 – Present

- Built an NLP-driven News Chatbot on dynamic LLM prompting workflows; wrote the internal technical guidelines for context-window budgeting under high-concurrency load.

FIMO Lab | *Research Intern (Geospatial AI)* 10/2025 – Present

- Built a Random Forest regression pipeline to process dense visual imagery at scale – the same frame-budget and memory discipline that carries directly into a real-time game loop.

FEATURED GAME DEV & AI PROJECTS

"AI-Boss" Web-Native Game Engine (Next.js 16 + React 19 + TypeScript) 03/2026 – Present

- Programmed a frame-rate-independent game loop on `requestAnimationFrame` with a hard 50ms delta-time cap to kill the spiral-of-death under CPU stalls; scaled velocity through a real-time `slowMotion` scalar for bullet-time without corrupting physics state, and added 50–120ms hit-stop freeze-frames to sell impact.
- Split collision into broad-phase distance culling and narrow-phase AABB overlap, decoupling hitbox geometry from entity transforms; resolved melee knockback as a configurable impulse vector with separate horizontal/vertical scaling, and drove homing missiles with kinematic seek interpolation for smooth pursuit curves.
- Architected a 7-state weighted-scoring FSM for the boss (idle, moving, melee, ranged, laser cast/beam, block, defeated) that samples attacks via roulette-wheel selection over distance, behavior persona, player height, and `iFrame` state, with injected noise to stop pattern-reading; layered on a reactive dodge/parry system and predictive ballistic aiming that leads the target using velocity, time-of-flight, and gravity drop.
- Dropped the OpenAI/Anthropic SDK wrappers to cut network overhead – called the API directly over `fetch` inside Next.js Server Actions to keep the key server-side, and truncated response streams to head/tail characters for fast structural sanity checks without flooding the terminal.

Google Summer of Code (GSoC 2026) — Contributor (OSGeo) 03/2026 – Present

- Built a Deep Learning framework for high-throughput spatial data validation; the continuous-stream data handling maps directly onto procedural map generation pipelines.

Trainify — Interactive AI Simulation & State Design 07/2025

- Prototyped camera tracking and simulation interaction flows in Figma, structured around the same state-transition logic that drives an NPC finite state machine.

EDUCATION

VNU University of Engineering and Technology (UET) 2024 – 2029 (Exp.)

- Bachelor of Engineering in Computer Engineering.
- **Key Coursework:** Systems Programming, Data Structures & Algorithms (Grade: A), Database Systems (SQL), Signals & Systems (Grade: B+).

SKILLS, HONORS & EXTRACURRICULARS

- **Technical Skills:** Prompt Engineering, LLM APIs (OpenAI, Claude, Gemini), Next.js 16, React 19, TypeScript, Tailwind CSS v4, Figma, n8n, C/C++, Linux Administration, Multithreading, SQL, HTML5 Canvas 2D.
- **Academic Honors:** **Top 15** UET IAI Hackathon 2025 (48-hour crunch production) | **Top 20** I-INVEST! 2026 | IAAC 2026 Award.
- **Extracurricular:** Weekly Winner, "Road to Mount Olympia" (VTV3), 2023 — fast retrieval and precision performance under time pressure.